

Davian Albarran

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WORK EXPERIENCE

Associate Full Stack Software Engineer

Jul 2024 — Present

MITRE

Shrewsbury, NJ

- Revamped four extensible Java plugins with well-documented APIs for the U.S. Army's Electronic Warfare Integration team, enabling seamless message transactions and mission tracking.
 - Enabled sending 3 new message types with 5 variants each.
- Spearheaded the creation of a novel, Dockerized dashboard system that visualizes signal detection events for the Army, enhancing situational awareness with user-friendly interfaces.
 - Constructed and deployed a Dockerized PostGIS database that enabled real-time geospatial data processing with a minimum 95% cache hit rate, improving situational awareness for SIGINT, COMMINT, and ELINT teams.
 - Configured and maintained a GeoServer Docker container to generate map tiles of geospatial data, supporting thousands of concurrent map tile requests and enhancing data visualization.
 - Influenced implementation of a prototype dashboard with Elasticsearch and Kibana Docker containers.
 - Developed a Vue web dashboard integrating with a GeoServer Docker container, enabling user interaction with geospatial data and adding the ability for users to submit their own reports.
 - Crafted developer tooling and build automations with Python and BASH scripts, streamlining development workflows and reducing build time by 50%, improving infrastructure efficiency.
 - Built a Go tool for running quick integration tests, giving developers a faster feedback loop before committing service changes.
- Coordinated with task leaders with estimating Agile story points for tasks, managing project timelines, and helped maintain the team Kanban board.

Full Stack Computer Science Intern

Jun 2022 — Aug 2023

MITRE

Eatontown, NJ

- Made key web development contributions for an FCC effort providing aid to the deaf and hard of hearing as a full stack engineer in an Agile environment.
- Implemented user profile picture feature, managing end-to-end development from backend storage in AWS S3 to frontend integration using vanilla JavaScript, enabling hundreds of agents and managers to personalize profiles.
- Investigated and developed a solution to stitch together WebRTC feeds for agent/caller recordings during a part-time assignment, improving recording completeness by 50%.
- Delivered front-end features using HTML, CSS, JavaScript, React and jQuery.
- Maintained and optimized backend endpoints using Node.js and Express, reducing average API latency by 10ms for the FCC application.
- Managed data storage across AWS S3, MongoDB, and MySQL, handling TBs of data to optimize user data, profile picture, and audio caption storage.
- Transitioned from full-time to part-time while continuing contributions during the academic year.

EDUCATION

Monmouth University

Sep 2020 — May 2024

Bachelor's of Science, Software Engineering w/ Minors in Mathematical & Computer Science

West Long Branch, NJ

- GPA: 3.74
- Awards: Dean's List, Dep. of Chemistry and Physics Physics Award, CSSE Award for Academic Excellence

PROJECTS

Owner/Maintainer, Ring Camera Metrics Tracker (<https://github.com/davianalbarran/ring-camera-metrics-tracker>) Jan 2026 — Present

- Architected end-to-end, containerized system using Python, Kafka, Go, Postgres, and Redis to track Ring camera metrics for my household of two users.
- Developed a Python ingestion service to read events from the Ring API, processing 45 events per day, and producing Kafka events.

- Engineered a Go consumer service to consume Kafka events and write to a Postgres database for durable, long-term event storage, capable of handling 10 TB of data.
- Implemented Redis in-memory storage, improving metric retrieval latency by 15x for efficient access to hot, low-latency metrics.

Owner/Maintainer, ParkShark (<https://github.com/MU-ParkShark/ParkSharkAPI>)

Nov 2023 — Present

- Lead architect of backend development of a cloud-native university parking management platform using AWS cloud services.
- Designed and maintained a 10,000+ line codebase in Node.js, Express, and Typescript.
- Collaborated cross-functionally with UI designers and a product stakeholder to define product requirements and deliver MVP features on schedule.
- **Note: the above URL is the public facing URL, but does not fully encapsulate the whole, up to date system.**

SKILLS

- **Programming Languages:** Java, JavaScript/TypeScript, HTML/CSS, Python, Bash, Kotlin, C/C++, Rust, Go
- **Technologies:** NextJS, ReactJS, tRPC, Node, Express, Fastify, NestJS, Docker, Kubernetes, Vue, PostgreSQL, PostGIS, MongoDB, Elasticsearch, Git, UNIX, Tailwind CSS, Bootstrap, NGINX, AWS Lambda, AWS EC2, REST, gRPC, GraphQL, Codex

EXTRACURRICULAR ACTIVITIES

Hackathons

Sep 2019 — Present

- Co-Founded a high school hack club, introducing competitive programming concepts to 15 students.
- Moderated a hackathon for 20 participants, providing technical support and ensuring adherence to time limits.
- Participated and won an internal company hackathon at MITRE focused on adding monitoring capabilities to our internal software wellness training platform for supervisors to be able to track their staffs completed trainings.
 - Developed monitoring capabilities by adding a database migration for supervisor roles, which would impact around 100 supervisors in our division.